

Ping Services: Reducing maintenance costs and powering renewable energy take-up with Google Cloud

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About Ping Services



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With Google Cloud, Ping Services enables operators to detect damage to turbine blades through real-time continuous monitoring of blade acoustics, saving A\$500 million annually.

Headquartered in South Australia, Ping Services currently collects acoustic data from turbines in Europe, USA, South America, and Australia and utilizes in-house software and algorithms to identify blade damage and assist wind farm operations. The technology can be adapted to assist other industries such as mining, rail, and water.

Industries: Technology, Energy,
Chemicals & Utilities

year across the w
industry.

Location: Australia

Google Cloud results

- Supports plans to scale to 250,000 installations and expand into new industries as an intelligent listening business
- Enables the business to access all the cloud products and services required without having to work with multiple providers
- Ensures leaders and

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Compute Engine

Google Cloud (<https://cloud.google.com/>)
Firestore (<https://cloud.google.com/firestore/>),

Cloud Storage (<https://cloud.google.com/storage/>)

Firebase Authentication

Cloud Functions

(<https://cloud.google.com/functions/>)

Compute Engine

(<https://cloud.google.com/compute/>)

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Cloud Firestore

(<https://cloud.google.com/firestore/>)

Firebase (<https://firebase.google.com/>)

Firebase Authentication

(<https://firebase.google.com/products/auth>)

**team
members can
focus on
developing
algorithms,
improving
reliability, and
doing a
better job for
clients rather
than on
administering
infrastructure**

The shift to renewable energy is gaining momentum worldwide and wind power is key to the transition. Wind is among the lowest cost ways of generating renewable energy at scale, with the sweeping blades of large turbines converting kinetic energy from wind into electricity exported to the grid. In Australia alone, wind supplied 35.9% of the clean energy and 9.9% of overall electricity used in 2020. However, the size of wind turbine blades and arcs that can reach 100 meters places immense pressure on turbine structures and components, potentially leading to failure, cracking, and erosion. Inclement weather, lightning strike, fire, and foreign objects can also contribute to blade damage.

Up to 23% of wind turbine failures are caused by blade damage, while 25%–30% of wind farm operation costs are related to repair and maintenance. The industry estimates about 3,800

blade failures occur per year at a total repair cost of about A\$3 billion.

Extending wind turbine blade life

By maintaining blades properly, wind farm operators can identify damage early and make minor repairs to extend blade life and avoid expensive unplanned downtime. If an operator repairs damage to a blade before it becomes severe, they can avoid spending hundreds of thousands of dollars on a replacement.

However, at several thousand dollars per tower, wind turbine blade maintenance is expensive, and so many operators schedule inspections only once every one to two years. The remote location of many wind turbine farms further increases the complexity of maintenance activities that can be difficult and dangerous with technicians required to inspect blades visually for damage before making repairs.

Founded in 2013, [Ping Services](https://pingmonitor.co/)

(<https://pingmonitor.co/>) is transforming blade maintenance through products that continuously analyze and monitor the acoustic signature of blades to detect damage. The signature of a damaged blade is different to that of an undamaged blade, due to changes in the sound of air moving over the surface. The business provides a compact solar-powered device that operators attach magnetically to the base of turbines to continuously “listen to” the blades as part of a solution that also uses algorithms to rate turbines’ blade health. Wind turbine operators can review and analyze results to

determine which turbines need maintenance via a simple app-based dashboard.



Challenged to listen

“We started eight years ago when a wind farm operator challenged us to create a product that could replicate his technicians’ approach of visiting turbines and listening for damage to the blades,” says Matthew Stead, Chief Executive Officer of Ping Services. The business’s founders worked on several iterations and prototypes of the solution to scope the requirement and determine whether the organization could cost-effectively access the computer or processing power needed to analyze the acoustic data captured.

Over Ping Services’ five years of fact-finding, computer power increased, costs decreased, and communication and Internet of Things technologies improved, making product development easier. Over the last three years, Ping Services has accelerated development and, with a device that incorporates its

own power supply and communications technology to send data from remote sites via cellular or satellite networks to a cloud-based platform and app, is targeting domestic and international markets. “We determined that cloud was the way to go, rather than assign scarce resources to maintaining software and servers, which we would have to do if we opted for on-premises equipment,” explains Jon Cooper, Chief Technology Officer of Ping Services.

“Google Cloud,
comprising
Cloud Storage,
Cloud Functions,
and Compute
Engine, and
Firebase,
comprising
Firebase
Hosting, Cloud
Firestore for
Functions, Cloud
Storage for
Firestore, and
Firebase
Authentication,

provided nice,
simple packages
with properly
documented
products and
services that
allowed us to
build everything
we needed to
without having
to go to multiple
providers.”

—Jon Cooper, Chief
Technology Officer, Ping
Services

Google Cloud and Firebase provide a one-stop shop

Ping Services’ evaluation found Google Cloud and Firebase provided all the products, services, and features the business needed in a single location, backed by the authentication and security needed to protect client data. “Google Cloud, comprising [Cloud Storage](#) (/storage), [Cloud Functions](#) (/functions), and [Compute Engine](#) (/compute), and Firebase, comprising [Firebase Hosting](#) (<https://firebase.google.com/docs/hosting>), [Cloud Firebase for Functions](#) (<https://firebase.google.com/docs/functions>), [Cloud](#)

Storage for Firebase

(<https://firebase.google.com/docs/storage>), and Firebase Authentication (<https://firebase.google.com/docs/auth>), provided nice, simple packages with properly documented products and services that allowed us to build everything we needed to without having to go to multiple providers,” says Jon. “Two of our skilled developers completed all the work needed to leverage these products without needing to call on Google experts or partners.”

In late 2020, Ping Services reviewed its architecture to determine whether Google Cloud was the best fit for the organization as it began targeting wind farm operators in Australia, the United States, and Europe with the latest commercially optimized version of its product. The organization evaluated global IoT cloud providers, reviewed analyst reports, and considered the latest IoT solutions available. “Through that evaluation process, we decided we would stay with Google Cloud, given the clarity provided by the interface and the efficiency of the command-line tools which make it easy to use,” says Jon.

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*—Matthew Stead, Chief
Executive Officer, Ping
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Helping clients meet cost requirements

With Google Cloud providing the infrastructure and resources that allow Ping Services to monitor and analyze blade acoustic signature from each turbine monitored, the business is helping clients reduce the cost of blade maintenance. “While wind turbines are very high-value assets, every dollar is important in the operations and maintenance phase,” explains Matthew. “With Google Cloud comprising less than 5% of our charge rates, we have the flexibility to help

clients meet their cost requirements.” Overall, the business believes it can save the wind energy industry A\$500 million in maintenance and repair costs.

Furthermore, while Ping Services styles itself as a hardware-as-a-service business, it is aiming for the agility and rapid development cycles of a software-as-a-service business to accommodate ongoing customer feedback. “Google Cloud enables us to meet the specific demands of operators of wind farms in different environments and conditions worldwide,” says Matthew.

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Aiming for 250,000 installations

The business has deployed 400+ systems to wind turbines and is aiming for 2,000 by the end of 2022, before targeting 10,000 and ultimately 25% of the 1 million turbines expected to be operating within the next decade—making scalability and multinational operations a key requirement of Google Cloud as its cloud service. However, Ping Services’ vision is not limited to wind turbines or to renewables.

“Google Cloud enables us to focus on our algorithms, improving our reliability and doing a better job for our clients,” explains Matthew. “While we’re supporting the transition to renewable energy

by supporting the adoption of wind power, our vision is to become an intelligent listening business that can reduce operating and maintenance costs across industries such as mining, rail, and water. We'll apply the same business model and adapt our technology to help organizations in these sectors improve maintenance and increase uptime—with Google Cloud as the foundation for our rollout.”

